

Powers of two as cycle lengths in graphs embedded in surfaces

Abstract

The Erdős–Gyárfás conjecture asserts that every finite simple graph of minimum degree at least 3 contains a cycle whose length is a power of two. We prove two new cases. First, every planar graph of minimum degree at least 4 contains a cycle of length 4 or 8; the same holds for graphs embeddable in the projective plane, and for 2-connected graphs embeddable in the torus or the Klein bottle. Second, every bipartite graph of minimum degree at least 3 embeddable in the torus contains a cycle of length 4, 8 or 16; in the planar case a 4-cycle already occurs. The engine is an extremal bound of independent interest: a connected graph of minimum degree at least 2 with no C_4 and no C_8 , 2-cell embedded in a surface S , satisfies $|E| \leq 2|V| - 2\chi(S)$. Specialised to the sphere this gives $\text{ex}_{\mathcal{P}}(n, \{C_4, C_8\}) \leq 2n - 4$, which appears to be the first planar Turán bound for a set of forbidden cycle lengths, and which we sharpen to $20|E| \leq 39|V| - 78$ under minimum degree 3. All bounds are shown to be attained or nearly attained by explicit graphs.

1 Introduction

All graphs are finite and simple. We write $n = |V(G)|$, $m = |E(G)|$, and $L(G)$ for the *cycle spectrum* of G , the set of lengths of its cycles. A *power cycle* is a cycle of length 2^k with $k \geq 2$.

Conjecture 1 (Erdős and Gyárfás, 1994). *Every graph with $\delta(G) \geq 3$ contains a power cycle.*

Conjecture 1 is open. Erdős later wrote that he had become convinced it is false, but no counterexample is known. The known proved cases are: $K_{1,m}$ -free graphs with $\delta \geq m + 1$, and graphs with $\Delta \geq 2m - 1$ (Shauger); planar claw-free graphs (Daniel and Shauger); 3-connected cubic planar graphs, which contain a 2^m -cycle for some $2 \leq m \leq 7$ (Heckman and Krakovski); graphs with no long induced path (Gao and Shan; Hu and Shen; Hegde, Sandeep and Shashank); and graphs of diameter 2 (Carr). A theorem of Liu and Montgomery implies the conjecture for all graphs of average degree exceeding an absolute constant, so the conjecture is entirely a bounded-degree question. On the computational side, Markström verified it for all cubic graphs on fewer than 29 vertices; Exoo constructed a cubic graph on 78 vertices with no cycle of length 4, 8 or 16, and one on 450 vertices with no cycle of length 4, 8, 16 or 32.

Notably, *no* case of Conjecture 1 with a minimum-degree hypothesis above 3 appears in the literature, and no case for planar graphs beyond the claw-free and 3-connected cubic ones. We supply both.

Theorem 2. *Every planar graph with $\delta(G) \geq 4$ contains a cycle of length 4 or 8. The same conclusion holds for every graph of Euler genus at most 1 with $\delta(G) \geq 4$, and for every 2-connected graph of Euler genus at most 2 with $\delta(G) \geq 4$.*

Theorem 3. *Every bipartite graph with $\delta(G) \geq 3$ that embeds in the torus contains a cycle of length 4, 8 or 16. In particular Conjecture 1 holds for bipartite graphs of orientable genus at most 1. Every planar bipartite graph with $\delta(G) \geq 3$ already contains a 4-cycle.*

Both theorems rest on the following extremal bound, which we believe is of independent interest.

Theorem 4. *Let G be a 2-connected simple graph with $n \geq 4$, containing no cycle of length 4 and no cycle of length 8, and 2-cell embedded in a surface S . Then*

$$m \leq 2n - 2\chi(S).$$

For the sphere this reads $m \leq 2n - 4$, i.e. $\exp(n, \{C_4, C_8\}) \leq 2n - 4$. The known planar Turán bounds for single cycles are $\exp(n, C_4) \leq \frac{15}{7}(n-2)$ (Dowden) and $\exp(n, C_8) \leq \frac{69}{25}(n-2)$ (Bai, Liu, Nie and Zhang); our bound is smaller than either, and coincides with the exact value $\exp(n, C_3) = 2n - 4$. We are not aware of any previous planar Turán bound for a set of two forbidden cycle lengths. Under the additional hypothesis $\delta(G) \geq 3$ we sharpen it to $20m \leq 39n - 78$ (Theorem 14).

The combinatorial heart of all of this is a short lemma with no topological content whatsoever.

Lemma 5 (Detour Lemma). *Let G contain no 4-cycle. If some 5-cycle of G has three of its edges lying in triangles, then G contains an 8-cycle.*

2 The Detour Lemma

Lemma 6. *In a C_4 -free graph every 5-cycle is chordless.*

Proof. A chord of a 5-cycle joins two vertices at distance 2 along the cycle, splitting it into arcs of lengths 2 and 3; the chord together with the arc of length 3 is a 4-cycle. $\square$

Lemma 7. *Let G be C_4 -free, let C be a 5-cycle of G , and let $uv \in E(C)$ lie in a triangle uvw . Then $w \notin V(C)$, and uv lies in exactly one triangle.*

Proof. Write $C = uvx_1x_2x_3$. Suppose $w \in V(C)$, so $w \in \{x_1, x_2, x_3\}$. On a 5-cycle each of x_1, x_2, x_3 is adjacent along C to at most one of u, v ; hence at least one of wu, wv is a chord of C , contradicting Lemma 6. If uv lay in two triangles with distinct apexes $w \neq w'$, then $ww'v$ would be a 4-cycle. $\square$

Lemma 8. *Let G be C_4 -free and let C be a 5-cycle. If $e \neq e'$ are edges of C lying in triangles with apexes $w, w' \notin V(C)$, then $w \neq w'$.*

Proof. Suppose $w = w'$ and write $C = v_1v_2v_3v_4v_5$. Up to the symmetry of C we may take $e = v_1v_2$ and $e' \in \{v_2v_3, v_3v_4, v_4v_5\}$.

If $e' = v_2v_3$ then w is adjacent to v_1, v_2, v_3 and $wv_1v_2v_3$ is a 4-cycle. If $e' = v_3v_4$ then w is adjacent to v_1, v_2, v_3, v_4 and $wv_4v_5v_1$ is a 4-cycle. If $e' = v_4v_5$ then w is adjacent to v_1, v_2, v_4, v_5 and $wv_2v_3v_4$ is a 4-cycle. In each case the four listed vertices are distinct, because $w \notin V(C)$. $\square$

Proof of Lemma 5. Let C be a 5-cycle with edges e_1, e_2, e_3 lying in triangles with apexes w_1, w_2, w_3 . By Lemma 7 the w_i lie off C , and by Lemma 8 they are pairwise distinct. Replace each $e_i = uv$ by the path $uw_i v$. The resulting closed walk visits the five vertices of C in cyclic order with w_1, w_2, w_3 inserted, so it visits eight pairwise distinct vertices, each once, and consecutive vertices are adjacent. Its length is $5 - 3 + 3 \cdot 2 = 8$. $\square$

Remark 9. Lemma 5 is equivalent to: in a $\{C_4, C_8\}$ -free graph, every 5-cycle has at most two edges lying in triangles. We verified it by exhaustive computation over all 143,038 connected C_4 -free graphs of order at most 11, of which 1,366 contain a 5-cycle with three edges in triangles; every one of those contains an 8-cycle.

3 The density theorem

We use standard facts about 2-cell embeddings. If G is 2-connected and 2-cell embedded in S , every face boundary is a cycle of G and every edge lies on exactly two distinct faces. Euler's formula reads $n - m + f = \chi(S)$, where f is the number of faces.

Proof of Theorem 4. Fix a 2-cell embedding. Since every face is bounded by a cycle and G has no 4-cycle and no 8-cycle, there are no faces of length 4 and none of length 8. Assign initial charges

$$\mu(v) = d(v) - 4 \quad (v \in V), \quad \mu(f) = \ell(f) - 4 \quad (f \text{ a face}).$$

Since $\sum_v d(v) = 2m$ and $\sum_f \ell(f) = 2m$, and $n + f = m + \chi(S)$,

$$\sum_v \mu(v) + \sum_f \mu(f) = (2m - 4n) + (2m - 4f) = 4m - 4(n + f) = -4\chi(S).$$

Discharging rule. For every edge e having a face of length 3 on one side and a face g on the other, g sends $\frac{1}{3}$ to that 3-face across e .

The rule is well defined. Each edge lies on two distinct faces, and no two 3-faces are adjacent: if triangles uvw and uvw' bounded the two faces at uv , then $w \neq w'$ (a face is determined by its boundary and its side) and $wuw'v$ would be a 4-cycle. Consequently a 3-face never sends charge, and each face g sends at most $\frac{1}{3}$ across each of its $\ell(g)$ edge-sides.

Let μ^* denote the final charges. Vertices neither send nor receive, so $\sum_v \mu^*(v) = 2m - 4n$. For faces:

- $\ell(f) = 3$: f receives $\frac{1}{3}$ across each of its three edges and sends nothing, so $\mu^*(f) = -1 + 1 = 0$.
- $\ell(f) = 4$ or $\ell(f) = 8$: no such face exists.
- $\ell(f) = 5$: the boundary is a 5-cycle. If three of its edges had 3-faces on the other side, those three edges would lie in triangles and Lemma 5 would produce an 8-cycle. Hence f sends at most $2 \cdot \frac{1}{3}$, and $\mu^*(f) \geq 1 - \frac{2}{3} = \frac{1}{3} > 0$.
- $\ell(f) = \ell \geq 6$: $\mu^*(f) \geq \ell - 4 - \frac{\ell}{3} = \frac{2\ell - 12}{3} \geq 0$.

Therefore $\sum_f \mu^*(f) \geq 0$, and

$$2m - 4n = \sum_v \mu^*(v) = -4\chi(S) - \sum_f \mu^*(f) \leq -4\chi(S),$$

which is the assertion. $\square$

Corollary 10. *Every connected planar graph with $\delta \geq 2$, $n \geq 4$ and no C_4 and no C_8 satisfies $m \leq 2n - 4$. In particular $\exp(n, \{C_4, C_8\}) \leq 2n - 4$.*

Proof. For 2-connected G this is Theorem 4 with $\chi = 2$, using that every embedding of a connected graph in the sphere is 2-cell. In general let $B_1, \dots, B_b$ be the blocks, with n_i, m_i vertices and edges. Then $\sum_i n_i = n + b - 1$ and $\sum_i m_i = m$. If $n_i \geq 4$ then B_i is 2-connected and $m_i \leq 2n_i - 4$; if $n_i = 3$ then B_i is a triangle and $m_i = 2n_i - 3$; if $n_i = 2$ then $m_i = 1 = 2n_i - 3$. In all cases $m_i \leq 2n_i - 3$, so $m \leq 2(n + b - 1) - 3b = 2n - 2 - b$. For $b \geq 2$ this gives $m \leq 2n - 4$, and for $b = 1$ the graph is 2-connected. $\square$

We verified Corollary 10 exhaustively over all 150,791 connected C_4 -free planar graphs with $\delta \geq 2$ and $4 \leq n \leq 13$, of which 8,242 are also C_8 -free.

4 Minimum degree four

Proof of Theorem 2. Suppose G has $\delta(G) \geq 4$, no C_4 and no C_8 . Passing to a component we may assume G connected. Since $\delta \geq 4$, no block of G is a single edge; let B be an endblock, that is, a block containing at most one cut vertex c of G . Then B is 2-connected, every $v \in V(B) \setminus \{c\}$ has $d_B(v) = d_G(v) \geq 4$, and $d_B(c) \geq 2$. Hence $|V(B)| \geq 5$ and

$$2|E(B)| \geq 4(|V(B)| - 1) + 2, \quad \text{i.e.} \quad |E(B)| \geq 2|V(B)| - 1.$$

B inherits $\{C_4, C_8\}$ -freeness and embeds in the same surfaces as G .

Euler genus 0 or 1. Embed B in a surface S of its own Euler genus $\gamma \leq 1$. If $\gamma = 0$ this is a sphere embedding of a connected graph, hence 2-cell. If $\gamma = 1$ then B is non-planar; a face of an embedding in the projective plane that is not a disc is a Möbius band, whose complement is a disc, which would make B planar; so the embedding is 2-cell. Now $\chi(S) = 2 - \gamma \geq 1$ and Theorem 4 gives $|E(B)| \leq 2|V(B)| - 2$, contradicting the display above.

Euler genus 2, G 2-connected. Take $B = G$ and fix a minimum-genus embedding, which is 2-cell. We may assume the Euler genus is exactly 2, so $\chi = 0$ and Theorem 4 gives $m \leq 2n$, while $\delta \geq 4$ gives $m \geq 2n$. Hence $m = 2n$ and every inequality in the proof of Theorem 4 is tight. Therefore $\sum_v (d(v) - 4) = 0$ with all degrees at least 4, so G is 4-regular; and $\mu^*(f) = 0$ for every face, so, since $\mu^* > 0$ for faces of length 5 and of length ≥ 7 , and no 4-faces exist, every face has length 3 or 6, and every 6-face sends the full $6 \cdot \frac{1}{3}$, i.e. all six of its edge-sides border a 3-face.

Every edge separates a 3-face from a 6-face: two 3-faces on one edge is excluded as before, and if both faces at an edge had length 6 then some 6-face would have an edge-side not bordering a 3-face.

Every 6-face is bounded by a 6-cycle. Its boundary walk does not backtrack (that would force a vertex of degree 1). If it is not a cycle it splits into two closed walks of lengths at least 3 summing to 6, so it has the form w, p, q, w, r, s with wpq a triangle. The edge wp has the

6-face on one side and a 3-face T' on the other. If T' is bounded by wpq then all three edges of wpq separate these two faces, forcing $\deg(p) = 2$, contradicting 4-regularity; otherwise T' is bounded by wpt with $t \neq q$, and the triangles wpq , wpt share wp , giving the 4-cycle $qwt p$.

Since consecutive faces around a vertex share an edge and every edge is of type $(3, 6)$, the faces around each vertex alternate 3, 6, 3, 6. Fix v with $N(v) = \{a, b, c, d\}$ in rotation and faces T_1 (between va, vb), H_1 (between vb, vc), T_2 (between vc, vd), H_2 (between vd, va), with $\ell(T_i) = 3$, $\ell(H_i) = 6$. Then T_1 is the triangle vab , so $ab \in E$; T_2 is the triangle vcd , so $cd \in E$; and H_1 is a 6-cycle $vbx y z c$. Consider the closed walk

$$W^* : \quad v, a, b, x, y, z, c, d, v,$$

all of whose consecutive pairs are edges of G . Its eight vertices are distinct: a, b, c, d are distinct and differ from v ; b, x, y, z, c lie on the 6-cycle H_1 and differ from v ; and

- $a = x$ would make ab the edge bx of H_1 , so T_1 and H_1 would both have bv, ba as their edges at b , forcing $\deg(b) = 2$;
- $a = y$ would make va a chord joining antipodal vertices of H_1 , splitting it into two 4-cycles;
- $a = z$ would give $a \sim c$, so vac and vab are triangles sharing va , giving the 4-cycle $b v c a$;
- $d = z, d = y, d = x$ are excluded symmetrically.

Hence W^* is an 8-cycle, a contradiction. □

Remark 11 (Sharpness). The hypothesis $\delta \geq 4$ cannot be weakened to $\delta \geq 3$: the truncated icosahedron (the buckyball) is a 3-connected cubic planar graph on 60 vertices with no C_4 and no C_8 , its smallest power cycle being a C_{16} . The conclusion cannot be strengthened by dropping either forbidden length: the icosidodecahedron is a 4-regular planar C_4 -free graph on 30 vertices, and it contains an 8-cycle, as Theorem 2 requires. Indeed the following holds.

Proposition 12. *Every planar graph with $\delta \geq 4$ and no C_4 has at least 30 vertices, and 30 is attained by the icosidodecahedron.*

Proof. Take an endblock B as above, so $|E(B)| \geq 2|V(B)| - 1$, and fix a plane embedding. All faces are bounded by cycles, there is no 4-face, and no two 3-faces share an edge, so each edge lies on at most one 3-face and $3f_3 \leq |E(B)|$. Hence

$$2|E(B)| = \sum_f \ell(f) \geq 3f_3 + 5(f - f_3) = 5f - 2f_3 \geq 5f - \frac{2}{3}|E(B)|,$$

so $f \leq \frac{8}{15}|E(B)|$, and Euler gives $|V(B)| = 2 + |E(B)| - f \geq 2 + \frac{7}{15}|E(B)|$, i.e. $|E(B)| \leq \frac{15}{7}(|V(B)| - 2)$. With $|E(B)| \geq 2|V(B)| - 1$ this yields $14|V(B)| - 7 \leq 15|V(B)| - 30$, so $|V(B)| \geq 23$. If G is 2-connected we may take $B = G$ and use $m \geq 2n$, giving $|V| \geq 30$; otherwise G has two endblocks sharing at most one vertex, so $n \geq 23 + 23 - 1 \geq 30$. □

5 A sharper planar bound

Lemma 13. *Let G be a 2-connected plane graph with $\delta(G) \geq 3$, no C_4 and no C_8 . For a face f let $t(f)$ be the number of edges of ∂f whose other face is a triangle. Then*

$$\ell(f) = 5 \Rightarrow t(f) \leq 2, \quad \ell(f) = 6 \Rightarrow t(f) \leq 1, \quad \ell(f) = 7 \Rightarrow t(f) = 0.$$

Proof. Write $C = \partial f$, a cycle of length $\ell \leq 7$, and let $e_1, \dots, e_t$ be the counted edges with triangle apexes $w_1, \dots, w_t$.

First, $w_i \notin V(C)$. For $\ell = 5$ this is Lemma 7. For $\ell \in \{6, 7\}$, write $C = uvx_1 \dots x_{\ell-2}$ with $e_i = uv$ and suppose $w_i = x_j$. If $j = 1$, then both uv and vx_1 lie on ∂f and on the triangle $T = uvx_1$, so the two faces at each of these edges are f and T ; a third edge at v would have to be drawn inside f or inside T , so $\deg(v) = 2$, contradicting $\delta \geq 3$. The case $j = \ell - 2$ is symmetric. For $2 \leq j \leq \ell - 3$ put $a = j$ and $b = \ell - 1 - j$, the two lengths of the C -paths from v to w_i and from w_i to u ; then $a, b \geq 2$ and $a + b = \ell - 1 \leq 6$, so $\min(a, b) \leq 3$. If $a = 2$ then uvx_1w_i is a 4-cycle; if $a = 3$ then $vx_1x_2w_i$ is a 4-cycle; and symmetrically for b .

Second, the w_i are pairwise distinct. For $\ell = 5$ this is Lemma 8. For $\ell \in \{6, 7\}$ suppose $w_i = w_j = w$. If e_i, e_j share a vertex, say $e_i = tu$ and $e_j = uv$ with $t \neq v$, then $twvu$ is a 4-cycle. Otherwise write $C = uvPstQu$ with $e_i = uv$, $e_j = st$, $|P| = p \geq 1$, $|Q| = q \geq 1$ and $p + q + 2 = \ell \leq 7$, so $\min(p, q) \leq 2$. If $p = 1$ then $uwsv$ is a 4-cycle; if $p = 2$, say vxs , then $vwsx$ is a 4-cycle; symmetrically for q .

Now replace any subset of $\{e_1, \dots, e_t\}$ by the corresponding detours as in Lemma 5: this produces cycles of every length in $[\ell, \ell + t]$. If $\ell = 5$ and $t \geq 3$, or $\ell = 6$ and $t \geq 2$, or $\ell = 7$ and $t \geq 1$, that interval contains 8. $\square$

Theorem 14. *Let G be a 2-connected plane graph with $\delta(G) \geq 3$, $n \geq 4$, no C_4 and no C_8 , and let f_3 be its number of triangular faces. Then*

$$7m \leq 13n + f_3 - 26, \quad \text{and hence} \quad 20m \leq 39n - 78.$$

Proof. Run the discharging of Theorem 4. By Lemma 13 the final charge of a face of length 5 is at least $1 - \frac{2}{3} = \frac{1}{3}$; of length 6 at least $2 - \frac{1}{3} = \frac{5}{3}$; of length 7 at least 3; and of length $\ell \geq 9$ at least $(2\ell - 12)/3 \geq 2$. There are no faces of length 4 or 8. Hence every non-triangular face retains at least $\frac{1}{3}$, so $\sum_f \mu^*(f) \geq (f - f_3)/3$. On the other hand $\sum_f \mu^*(f) = \sum_f \mu(f) = -8 - (2m - 4n) = 4n - 2m - 8$. With $f = m - n + 2$ this gives

$$4n - 2m - 8 \geq \frac{1}{3}(m - n + 2 - f_3),$$

i.e. $12n - 6m - 24 \geq m - n + 2 - f_3$, i.e. $7m \leq 13n + f_3 - 26$. Finally, C_4 -freeness makes triangles pairwise edge-disjoint, so $3f_3 \leq m$, and substituting gives $21m \leq 39n + m - 78$. $\square$

Corollary 15. *Every 2-connected planar graph with $\delta \geq 3$ and $20m \geq 39n - 77$ contains a cycle of length 4 or 8. Equivalently, writing $D = \sum_{d(v) \geq 4} (d(v) - 3) = 2m - 3n$, this holds whenever $D \geq (9n - 77)/10$. In particular, every 2-connected planar graph all of whose degrees are 3 or 4 and which has at most $(n + 77)/10$ vertices of degree 3 contains a cycle of length 4 or 8.*

Taking no vertices of degree 3 recovers the planar case of Theorem 2, so this is a strict extension. The constant in Theorem 14 cannot be pushed below 90/59: chaining k copies of the buckyball at cut vertices produces planar $\{C_4, C_8\}$ -free graphs with $\delta \geq 3$, $n = 59k + 1$ and $m = 90k$.

Remark 16. The same discharging yields a further structural restriction which may be useful elsewhere. If F is a set of faces of a 2-connected plane graph whose closed union is a disc, then ∂F is a cycle of length $\sum_{f \in F} \ell(f) - 2e_F$, where e_F counts edges incident with two faces of F . Applying this to two faces sharing exactly one edge shows that in a $\{C_4, C_8\}$ -free plane graph no two 3-faces are adjacent, *no two 5-faces are adjacent*, and no 3-face is adjacent to a 7-face.

6 Bipartite graphs of genus at most one

Proof of Theorem 3. Let G be bipartite with $\delta(G) \geq 3$, embeddable in the torus, and suppose G has no C_4 ; being bipartite, G then has girth at least 6. Passing to a component and then to an endblock B with cut vertex c (or $B = G$), we have B 2-connected with $d_B(v) = d_G(v) \geq 3$ for $v \neq c$ and $d_B(c) \geq 2$, so

$$2m \geq 3n - 1,$$

where from now on $n = |V(B)|$ and $m = |E(B)|$. Fix a minimum-genus embedding of B , which is 2-cell. Since B is 2-connected every face is bounded by a cycle, necessarily of even length at least 6; hence $2m = \sum_f \ell(f) \geq 6f$, i.e. $f \leq m/3$.

Genus 0. Euler gives $f = m - n + 2 \leq m/3$, so $m \leq \frac{3}{2}(n - 2)$ and $2m \leq 3n - 6 < 3n - 1$, a contradiction. Hence every planar bipartite graph with $\delta \geq 3$ contains a 4-cycle.

Genus 1. Euler gives $f = m - n \leq m/3$, so $2m \leq 3n$, and therefore $3n - 1 \leq 2m \leq 3n$. If n is odd then $3n$ is odd, so $2m = 3n - 1$ and $f = (n - 1)/2$; then $\sum_f \ell(f) = 2m = 3n - 1$ while $6f = 3n - 3$, and since all face lengths are even and at least 6, exactly one face has length 8, giving an 8-cycle. If n is even then $2m = 3n$, so B is cubic, $f = n/2$ and $\sum_f \ell(f) = 6f$: every face is a hexagon.

So B is a hexagonal tessellation of the torus. Its universal cover is the hexagonal tessellation $\mathcal{H}$ of the plane, and the deck group is a rank-2 lattice Λ' of translations, so $B = \mathcal{H}/\Lambda'$. Identify the translation lattice of $\mathcal{H}$ with $\mathbb{Z}^2$ and let $|\cdot|$ be the triangular norm, $|(a, b)| = |a| + |b|$ if $ab \geq 0$ and $\max(|a|, |b|)$ otherwise; two vertices of $\mathcal{H}$ have the same image in B exactly when they have the same type and their difference lies in Λ' . Put

$$t = \min\{|\lambda| : \lambda \in \Lambda' \setminus \{0\}\}.$$

Then $t \leq 1$ makes the quotient non-simple; $t = 2$ produces a 4-cycle; $t = 3$ produces an 8-cycle; and $t \geq 4$ produces a 16-cycle. For the last, take the polyhex whose four hexagon centres are $0, e_1, 2e_1, e_2$: it has four hexagons and four adjacent pairs, so its boundary has length $6 \cdot 4 - 2 \cdot 4 = 16$, and every difference of two same-type vertices along that boundary has norm at most 3, so the projection to B is injective when $t \geq 4$. For $t \in \{2, 3, 4\}$ and each of the 12, 18, respectively 24 lattice vectors λ of that norm we exhibit a path of length $2t$ in $\mathcal{H}$ from a vertex to its λ -translate all of whose same-type differences have norm smaller than t ; its projection is then a $2t$ -cycle. This is a finite verification which we carried out exhaustively, with no exceptions. $\square$

Remark 17 (Sharpness). All three lengths in Theorem 3 are needed. The hexagonal torus $\mathcal{H}/\Lambda'$ with $t = 5$ of smallest index, on 38 vertices, is a cubic bipartite graph of girth exactly 6 with cycle counts 19, 0, 171, 323, 969 at lengths 6, 8, 10, 12, 14: it has neither a 4-cycle nor an 8-cycle, and its smallest power cycle is a 16-cycle. Since $t \geq 5$ forces the triangular balls of radius 2 about lattice points to be disjoint, and each contains 19 points, the index of Λ' is at

least 19; hence 38 is exactly the minimum order of a toroidal bipartite graph with $\delta \geq 3$, girth 6 and no 8-cycle. In particular the implication “bipartite, $\delta \geq 3$, girth 6 $\Rightarrow C_8$ ” is false.

Remark 18. The Klein bottle and the projective plane are not covered by Theorem 3: the counting above already forces Euler genus at least 2 in the relevant range, and the classification of hexagonal tessellations we use is specific to the torus.

7 Concluding remarks

Theorem 4 degenerates exactly at $\chi = 0$, where the bound $m \leq 2n$ meets the hypothesis $m \geq 2n$ supplied by $\delta \geq 4$; this is why the torus and Klein bottle require the separate argument in the proof of Theorem 2, and why surfaces of negative Euler characteristic are out of reach by this method. For $\delta \geq 5$ the same counting gives $n \leq -4\chi(S)$, so for each fixed surface only finitely many orders survive.

The planar case of Conjecture 1 at $\delta = 3$ remains open, and two obstructions delimit it. First, one cannot hope to prove that a planar counterexample is cubic: identifying a vertex of one buckyball with a vertex of another yields a planar graph with $\delta \geq 3$, a vertex of degree 6, and no C_4 and no C_8 . Second, one cannot hope to conclude with the lengths 4, 8 and 16 alone: the buckyball is triangle-free and the truncated dodecahedron is pentagon-free, both are 3-connected cubic planar with no C_4 and no C_8 , and every 2-connected plane graph with $\delta \geq 3$ and no 4-face satisfies $\sum_f (6 - \ell(f)) \geq 12$ and therefore has a 3-face or a 5-face. This is consistent with the range $2 \leq m \leq 7$ in the theorem of Heckman and Krakovski, and with Exoo’s 3-connected cubic planar graph on 420 vertices having no cycle of length 2^m for $m \leq 4$.

Question 19. Is $\exp(n, \{C_4, C_8\}) = (\frac{3}{2} + o(1))n$? The constructions above give $\frac{90}{59}n + O(1)$ from below, and Theorem 14 gives $\frac{39}{20}n$ from above under $\delta \geq 3$.